

Evaluate MCP for Your AI Integration Scenario

You've explored MCP's benefits, studied industry success stories, and learned the evaluation framework. Now it's time to apply this knowledge to a real-world scenario that mirrors the challenges you might face in your own organization.

Objective

The goal isn't just to compare technologies—it's to apply the MCP evaluation framework to a real-world healthcare integration challenge and produce a decision-ready recommendation. Are you clearly assessing integration complexity, security compliance, team readiness, and ROI? Are you building an implementation strategy that addresses stakeholder concerns and supports secure, scalable, and efficient data interoperability?

Scenario

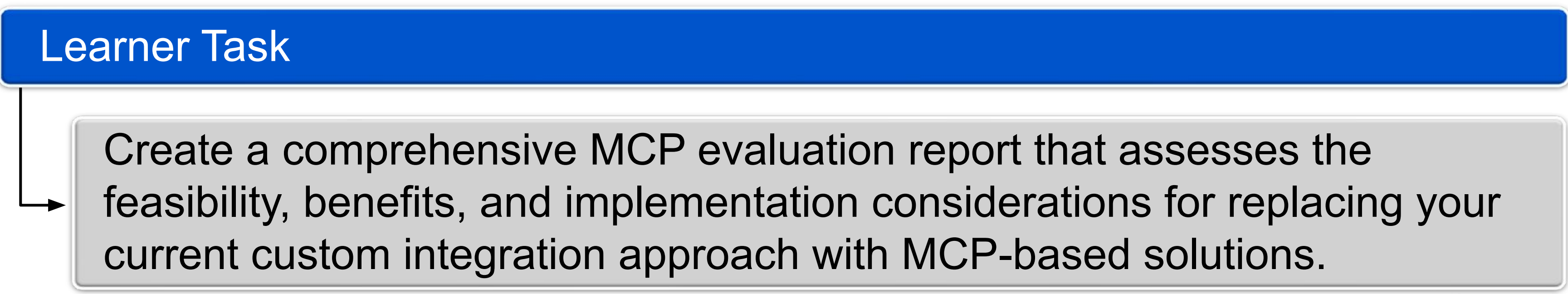
You're a senior AI architect at MedTech Solutions, a healthcare technology company building an AI-powered clinical decision support system. Your system needs to integrate with multiple data sources Electronic Health Records (EHR), lab systems, pharmaceutical databases, medical imaging systems, and real-time patient monitoring devices.

Currently, your team has built custom connectors for each system, but maintenance is becoming overwhelming. Security audits are increasingly complex, and each new integration takes weeks to develop and test. Your CTO has asked you to evaluate whether MCP could solve these integration challenges.

Your stakeholders include the Chief Technology Officer (urgency decision needed in 2 weeks), the Security team (concerned about maintaining HIPAA compliance), and the Development team (worried about learning new protocols). The clinical team needs reliable, real-time access to patient data to make life-critical decisions.

Instructions for learners

Learner Task



Create a comprehensive MCP evaluation report that assesses the feasibility, benefits, and implementation considerations for replacing your current custom integration approach with MCP-based solutions.

Refine the learner's task with completion criteria

Your evaluation report should include the following components

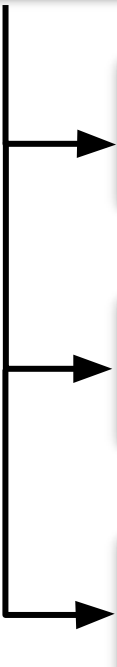
1. Integration Complexity Assessment

→ List the 5 data sources your AI system needs to connect with

→ Evaluate the complexity of each integration using the framework from the lesson

→ Assign a complexity score (1-5) for each data source

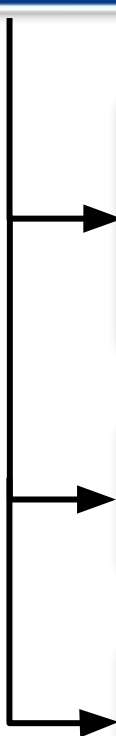
2. Security Requirements Analysis

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- Identify the key security requirements for healthcare data integration
 - Evaluate how MCP's security model addresses these requirements
 - Highlight any security gaps or concerns

3. Team Capability and Resource Evaluation

- Assess your team's readiness for MCP implementation
- Estimate the learning curve and training requirements
- Calculate the resource investment needed for MCP adoption

4. Cost-Benefit Analysis

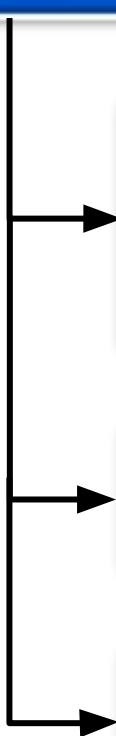


Compare the costs of maintaining current custom integrations vs. MCP implementation

Estimate the timeline for achieving ROI with MCP

Consider both short-term costs and long-term benefits

5. Recommendation and Implementation Strategy



Provide a clear recommendation (adopt MCP, continue current approach, or hybrid strategy)

Outline a phased implementation plan if recommending MCP

Address stakeholder concerns and mitigation strategies

Completion Criteria

- Five data sources identified with complexity scores
- Security analysis addressing healthcare compliance requirements
- Team capability assessment with training recommendations
- Cost-benefit analysis with ROI timeline
- Clear recommendation with implementation strategy
- Professional report format suitable for executive presentation

Submission

Submission Guidelines



Submit your evaluation report as a PDF, Word document, or structured outline. The report should be 800-1200 words and suitable for presentation to technical and non-technical stakeholders.

Reflection

Engage in reflection Transition to Dialogue



After completing your evaluation, you'll engage in a structured dialogue to test your analysis against alternative scenarios and refine your recommendation based on changing requirements and constraints.